

AIT2

Organic Chemistry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a first-year lab that uses ball-and-stick kits from a school supplier, a student builds ethene, C_2H_4 , where the two carbons share a double bond and each carbon also holds two hydrogens. The kit has 60 pieces. What is the hybridization of each carbon and the approximate H-C-H angle?

- A. sp^3 , about 109.5°
- B. sp^2 , about 109.5°
- C. sp , 180°
- D. sp^2 , about 120°

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A carbon atom has a 180° angle between its two neighbors. Which bonding pattern fits that carbon?

- A. It can only happen in carbon dioxide, never in a hydrocarbon.
- B. Four single bonds.
- C. One triple bond and one single bond.
- D. One double bond and two single bonds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A chemist developing a vanilla-scented candle line studies a compound drawn as $CH_3-CH_2-C(=O)-CH_3$. The carbonyl carbon is bonded to two other carbons. The supplier ships it in 5-gallon drums. To which functional group family does it belong?

- A. Ketone
- B. Carboxylic acid
- C. Ester, because fragrance compounds are esters.
- D. Aldehyde

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A gas supplier that serves 40 restaurants stores butane and 2-methylpropane side by side. Both have the formula C_4H_{10} ; butane is a straight chain and 2-methylpropane is branched. What is their relationship?

- A. The same compound, because the formula is the same.
- B. Structural (constitutional) isomers.
- C. Isotopes.
- D. Stereoisomers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Two molecules have the same formula and the same atom-to-atom connections, but differ in how groups sit on either side of a $C=C$ double bond. What are they?

- A. Conformers that interconvert by rotating the double bond.
- B. Cis-trans (geometric) stereoisomers.
- C. Structural isomers.
- D. Always the identical compound, since formula and connections match.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A drug developer reviews 2-butanol, $CH_3-CH(OH)-CH_2-CH_3$, a compound first made in the 1920s that sells for about \$3 per gram. Numbering from the left, which carbon is a chiral center?

- A. Carbon 3
- B. Every carbon bonded to oxygen, and every carbon next to one.
- C. Carbon 2
- D. Carbon 1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A solvent distributor with a warehouse near the port lists three compounds of nearly the same molar mass: propanol ($\text{C}_3\text{H}_7\text{OH}$, 60 g/mol), acetone (58 g/mol), and butane (58 g/mol). Which has the highest boiling point?

- A. Propanol
- B. All about the same, because molar masses match.
- C. Butane
- D. Acetone

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Pentane boils at 36 °C and 2,2-dimethylpropane boils at 10 °C. Both are C_5H_{12} . Why does the branched one boil lower?

- A. Its covalent bonds are weaker, so they break more easily.
- B. It has a lower molar mass.
- C. It forms hydrogen bonds that pull it apart.
- D. Its compact shape gives less surface contact and weaker London dispersion forces.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A quality-control analyst at a plant that runs three shifts receives an unknown hydrocarbon with formula C_5H_8 . Spectra show it has no rings. How many pi bonds does it contain?

- A. 0, because it has no rings.
- B. 4
- C. 2
- D. 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A hydrocarbon has 6 carbons, one ring, and no double or triple bonds. How many hydrogens does it have?

- A. 14
- B. 12
- C. 10
- D. 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A technician relabeling bottles for a college stockroom that reorganized its shelves last summer finds $\text{CH}_3\text{-CH}_2\text{-CH}(\text{CH}_3)\text{-CH}_2\text{-CH}_2\text{-CH}_3$. What is the correct IUPAC name?

- A. Heptane
- B. 3-methylpentane
- C. 4-methylhexane
- D. 3-methylhexane

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

At a plant that ships 20 tons of product a month, propene ($\text{CH}_2=\text{CH-CH}_3$) reacts with HBr with no peroxides present. What is the major product?

- A. 2-bromopropane
- B. A bromopropene, since the double bond stays.
- C. 1-bromopropane
- D. 1,2-dibromopropane

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

In a first-year lab ~~that uses ball-and-stick kits from a school supplier~~, a student builds ethene, C_2H_4 , where the two carbons share a double bond and each carbon also holds two hydrogens. ~~The kit has 60 pieces.~~ What is the hybridization of each carbon and the approximate H-C-H angle?

Target word: hybridization and angle at a double-bonded carbon **Decides:** Each carbon has one double bond and three electron domains.

A	Adjacent category	Counts four bonds and ignores that one of them is a double bond.
B	Dropped element	Right hybridization, but drops the angle that goes with it.
C	Confusion set	That geometry belongs to a triple-bonded carbon such as in ethyne.
D	Correct	Three electron domains (one double, two single bonds) give trigonal planar sp^2 geometry.

2. Answer C Concept 1 • transfer

A carbon atom has a 180° angle between its two neighbors. Which bonding pattern fits that carbon?

Target word: reverse from angle to bonding **Decides:** 180° means two electron domains, sp .

A	Absolute language	Alkynes such as ethyne have linear carbons.
B	Inverted operation	Four single bonds give 109.5° , the opposite end of the scale.
C	Correct	Two electron domains give a linear sp carbon.
D	Adjacent category	That is trigonal planar, 120° .

3. Answer A Concept 2

A chemist ~~developing a vanilla-scented candle line~~ studies a compound drawn as $CH_3-CH_2-C(=O)-CH_3$. The carbonyl carbon is bonded to two other carbons. ~~The supplier ships it in 5-gallon drums.~~ To which functional group family does it belong?

Target word: functional group **Decides:** $C=O$ bonded to two carbons inside the chain.

A	Correct	A carbonyl with carbon on both sides is a ketone (butanone).
B	Dropped element	Needs an OH on the carbonyl carbon, which is missing here.
C	Camouflage	Answers the scent detail; an ester needs $C=O$ bonded to an O-C.
D	Confusion set	An aldehyde carbonyl sits at the chain end and carries an H.

4. Answer B Concept 3

A gas supplier ~~that serves 40 restaurants~~ stores butane and 2-methylpropane side by side. Both have the formula C_4H_{10} ; butane is a straight chain and 2-methylpropane is branched. What is their relationship?

Target word: type of isomer **Decides:** Same formula, different connectivity.

A	Dropped element	Ignores connectivity, which differs.
B	Correct	Atoms are connected in a different order.
C	Confusion set	Isotopes are atoms with different neutron counts, not molecules.
D	Adjacent category	Stereoisomers share connectivity; these do not.

5. Answer B Concept 3 · transfer

Two molecules have the same formula and the same atom-to-atom connections, but differ in how groups sit on either side of a C=C double bond. What are they?

Target word: reverse: same connectivity, different arrangement **Decides:** Only spatial arrangement differs.

A	Confusion set	Double bonds do not rotate freely at room temperature.
B	Correct	Same connectivity, different fixed arrangement around a double bond.
C	Inverted operation	Reverses which property differs; connectivity is the same here.
D	Absolute language	Stereoisomers can differ in properties despite matching connectivity.

6. Answer C Concept 4

A drug developer reviews 2-butanol, $\text{CH}_3\text{-CH(OH)-CH}_2\text{-CH}_3$, ~~a compound first made in the 1920s that sells for about \$3 per gram~~. Numbering from the left, which carbon is a chiral center?

Target word: locate the chiral carbon **Decides:** C2 bears H, OH, CH₃, and CH₂CH₃ — four different groups.

A	Adjacent category	C3 has two hydrogens, so two groups repeat.
B	Absolute language	Chirality depends on four different groups, not on oxygen.
C	Correct	Four different groups make C2 a stereocenter.
D	Confusion set	C1 carries three identical hydrogens.

7. Answer A Concept 5

A solvent distributor ~~with a warehouse near the port~~ lists three compounds of nearly the same molar mass: propanol ($\text{C}_3\text{H}_7\text{OH}$, 60 g/mol), acetone (58 g/mol), and butane (58 g/mol). Which has the highest boiling point?

Target word: highest boiling point **Decides:** Only propanol has an O-H that can hydrogen bond.

A	Correct	Hydrogen bonding is the strongest intermolecular force here.
B	Dropped element	Drops the type of intermolecular force, which dominates at equal mass.
C	Inverted operation	Nonpolar, only London forces; it has the lowest boiling point.
D	Adjacent category	Polar with dipole-dipole forces, but no O-H to hydrogen bond.

8. Answer D Concept 5 · transfer

Pentane boils at 36 °C and 2,2-dimethylpropane boils at 10 °C. Both are C_5H_{12} . Why does the branched one boil lower?

Target word: explain the lower boiling point **Decides:** Branching reduces surface contact.

A	Adjacent category	Boiling separates molecules; it does not break covalent bonds.
B	Wrong stem number	Both are C_5H_{12} with the same mass.
C	Confusion set	Hydrocarbons do not hydrogen bond, and hydrogen bonds raise boiling points.
D	Correct	Less contact area means weaker dispersion forces between molecules.

9. Answer C Concept 6

A quality-control analyst ~~at a plant that runs three shifts~~ receives an unknown hydrocarbon with formula C_5H_8 . Spectra show it has no rings. How many pi bonds does it contain?

Target word: pi bonds from a formula **Decides:** $(2 \times 5 + 2 - 8) / 2 = 2$, and no rings.

A	Adjacent category	Conflates rings with all unsaturation; pi bonds also count.
B	Dropped element	$12 - 8 = 4$ without dividing by 2.
C	Correct	Two degrees of unsaturation, none used by rings, so two pi bonds.
D	Camouflage	Borrows the three shifts; the formula decides.

10. Answer B Concept 6 • transfer

A hydrocarbon has 6 carbons, one ring, and no double or triple bonds. How many hydrogens does it have?

Target word: solve for H **Decides:** One degree of unsaturation removes 2 H from $2(6) + 2 = 14$.

A	Dropped element	The open-chain alkane count; ignores the ring.
B	Correct	$14 - 2 = 12$ (cyclohexane, C_6H_{12}).
C	Inverted operation	Removes 4 H, as if the ring counted twice.
D	Confusion set	That is benzene, which also has three pi bonds.

11. Answer D Concept 7

A technician relabeling bottles for a college stockroom ~~that reorganized its shelves last summer~~ finds $CH_3-CH_2-CH(CH_3)-CH_2-CH_2-CH_3$. What is the correct IUPAC name?

Target word: IUPAC name **Decides:** Longest chain = 6 carbons; methyl gets the lower locant from the left end.

A	Confusion set	Same formula, C_7H_{16} , but heptane is unbranched.
B	Dropped element	Miscounts the chain, dropping a carbon.
C	Inverted operation	Numbers from the far end, giving a higher locant.
D	Correct	Six-carbon chain, methyl on C3 when numbered from the nearer end.

12. Answer A Concept 8

~~At a plant that ships 20 tons of product a month,~~ propene ($CH_2=CH-CH_3$) reacts with HBr with no peroxides present. What is the major product?

Target word: major addition product **Decides:** No peroxides, so Markovnikov: Br goes to the more substituted carbon.

A	Correct	H adds to CH_2 , Br to the middle carbon, via the more stable secondary carbocation.
B	Adjacent category	Describes substitution; alkenes undergo addition, which uses up the double bond.
C	Inverted operation	Anti-Markovnikov product, which needs peroxides.
D	Confusion set	That comes from adding Br_2 , not HBr.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Hybridization and geometry	Four single bonds = sp^3 , about 109.5° ; one double bond = sp^2 , about 120° ; a triple bond (or two doubles) = sp , 180° .	Calling every carbon sp^3 because carbon always makes four bonds.	A double or triple bond drawn at the carbon in question, or a bond angle given.
2	Functional group recognition	The group decides the family: $C=O$ inside a chain is a ketone, $C=O$ at the chain end with H is an aldehyde, $C=O$ with OH is a carboxylic acid, $C=O$ with O-C is an ester.	Naming the family from a smell, a use, or a name ending instead of the atoms.	Where the carbonyl sits and what is bonded to it.
3	Structural vs. stereoisomers	Isomers share a molecular formula; structural isomers differ in connectivity, stereoisomers share connectivity and differ only in 3-D arrangement.	Assuming the same formula means the same compound.	Two names or drawings with the same formula.
4	Chiral centers	A carbon is a chiral center when it bears four different groups.	Treating any carbon bonded to a heteroatom as chiral.	"Which carbon is chiral / stereocenter?"
5	Intermolecular forces and boiling point	At similar molar mass, hydrogen bonding > dipole-dipole > London dispersion; more surface area (less branching) raises dispersion forces.	Explaining boiling point with the strength of covalent bonds inside the molecule.	Compounds of similar mass ranked by boiling point.
6	Degree of unsaturation	Degrees of unsaturation = $(2C + 2 - H) / 2$; each ring or pi bond counts as one.	Forgetting to halve, or forgetting that a ring counts.	A molecular formula plus "how many rings or pi bonds".
7	IUPAC naming	Name from the longest continuous chain, numbered from the end that gives substituents the lowest locants.	Counting the straight line as drawn, or numbering from the wrong end.	A branched chain drawn in a line.
8	Markovnikov addition	When HX adds to an unsymmetrical alkene, H goes to the carbon that already has more H; X goes to the more substituted carbon (unless peroxides are present).	Putting the halogen on the end carbon.	An alkene plus HBr or HCl.

Course scope (public description): Organic Chemistry focuses on the study of compounds that contain carbon, much of which is learning how to organize and group organic compounds in order to predict their structure, behavior, and reactivity based on common bonds found within an organic compound.